**Aliah University**

End Semester Examination, 2023-2024

Course: Civil Engineering Year: Final Year Subject: Environmental Geo-technology

Subject Code: CENUGPE06 (PEC-VI) Max. Marks: 80 Times: 3 hrs

Answer any Eight Questions.

1. Why soil becomes important in Geo-environmental Engineering? With examples, discuss the multiphase behavior of soil. **(10 Marks)**
2. What is adsorption? Differentiate between specific and non-specific adsorption. List the forces responsible for adsorption reaction. **(2+3+5 Marks)**
3. Explain the significance of cation exchange capacity. How does CEC influence reactivity of soil? **(5+5 Marks)**
4. What is the aim of integrated solid waste management program? Differentiate between a natural attenuation landfill and an engineered landfill. **(6+4 Marks)**
5. Explain the important details required for deciding landfill site. **(10 Marks)**
6. What are the negative consequences of soil pollution? **(10 Marks)**
7. What are the processes involved in the planning of contaminated site remediation? **(10 Marks)**
8. Discuss the important physico-chemical methods for performing contaminated soil remediation. **(10 Marks)**
9. Discuss the important properties of clay minerals. **(10 Marks)**
10. Define sorption. What are the specific cases of sorption? Discuss the factors affecting sorption. **(2+3+5 Marks)**
11. Discuss the advantage and disadvantage of sanitary landfill. **(10 Marks)**

feasibility study

**Even Semester Examination 2023-24**

**Subject Code: CENUGPC24**

**Subject Name: Construction Management**

**B. Tech Civil Engineering: $8^{th}$ Semester**

**Full Marks: 80 (8x10)** **Time: 3 Hrs.**

*Attempt Any 8 Questions*

1. Define Program Evaluation Review Technique (PERT), and Critical Path Method (CPM). Write down the basic steps in PERT and CPM.
2. Explain the following terms:
   (a) Earliest Time (b) Latest Time (c) Slack (d) Critical Path
   (e) Most Likely Time
3. Prepare a rough cost estimate with following details:
   (a) Plinth Area $500m^2$ (b) Height of each storey=3m
   (c) No. of storey=G+3 (d) Cubical content rate=Rs.3500/$m^3$
   (e) Water supply 8% (f) Electricity=7%
   (g) Contingencies=2% (h) Contractor Profit=10%
4. Define long and short wall method with figure. Compute the volume of cement, sand, and coarse aggregate having total volume of 12cum, and ratio 1:2:4.
5. What are the rules for drawing network diagram? Also mention the common errors that occur in drawing networks.
6. A project has different characteristics:

| Activity | 1-2 | 2-3 | 2-4 | 3-5 | 4-5 | 4-6 | 5-7 | 6-7 | 7-8 | 7-9 | 8-10 | 9-10 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Optimistic Time | 1 | 1 | 1 | 5 | 4 | 5 | 6 | 8 | 4 | 7 | 1 | 5 |
| Pessimistic Time | 7 | 5 | 7 | 7 | 6 | 9 | 8 | 10 | 8 | 10 | 5 | 9 |
| Most Likely Time | 3.5 | 4 | 5 | 6 | 5 | 7 | 7 | 9 | 6 | 8 | 4 | 7 |

Construct Program Evaluation Review Technique (PERT) network. Also compute critical path, and variance of each event.

7. What are the different machineries used for construction? For a given activity, the optimistic time, pessimistic time and most probable estimates are 4,16, and 7 days respectively. Determine the expected time.
8. The time estimates obtain from three contractors P, Q and R for executing a particular job as under:

| Contractor | Optimistic Time | Most likely time | Pessimistic time |
|---|---|---|---|
| P | 8 | 13 | 16 |
| Q | 6 | 12 | 14 |
| R | 7 | 11 | 15 |

Which contractor is more certain about their job?

9. What are the different types of distribution? Why PERT always follow the β-distribution?

1-2 2-3 2-4 3-5 4-5 4-6 5-7 6-7 7-8 7-9 8-10 9-10
3.67 3.67 4.67 6 5 7 7 9 6 8.17 3.67 7

Comp

## Even (Spring) Semester Examination 2023-2024
## Paper Code: MBAUGHU02; Paper name: Professional Values and Ethics
### VIIIth Semester
### Full Marks: 80; Time: 3Hrs.

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*(The figures in the margin indicate full marks.*
*Candidates are required to give their answers in their own words as far as possible)*

**GROUP: A** (*Answer all the questions*) **(1 x 10 = 10)**

**1. Choose and write ONLY the correct option (a/b/c/d). DO NOT write full sentences.**

I. The rule of ethics is also called as
A. Rule  B. Law  C. Responsibility  D. None of the above

II. Ethical issues that can affect an Engineer's professional and personal life are termed as
A. Macro-ethics  B. Micro-ethics  C. Morals  D. Rights

III. Business malpractice does not include
A. Black Marketing  B. Advertisement  C. Duplication  D. Adulteration

IV. Dowry deaths, wife battering is an example of
A. Criminal violence  B. Domestic violence  C. Social violence  D. Gross violence

V. Ethics is the science of
A. beauty  B. conduct  C. truth  D. mind

VI. Value is
A. response of the society  B. enduring belief  C. material culture  D. non-material culture

VII. The word 'ethics' was derived from the Greek word
A. ethies  B. ethos  C. ethees  D. ethise

VIII. Aesthetics deals with the standard of
A. truth  B. beauty  C. goodness  D. trust

IX. Business ethics has a ____________ application.
A. natural  B. universal  C. practical  D. none of the above

X. The relevance of ethics is in its
A. Context  B. Applications  C. Principles  D. Understanding

**GROUP: B** (*Answer any five questions*) **(5 x 5 = 25)**

2. What do you mean by 'Values' and 'Ethics'?
3. What are the features of 'Rapid Technology growth'? Discuss.
4. Write a short note on 'Solar Energy'.
5. What are the problems of Man machine Interactions?
6. Write a short note on 'Canons of Ethics'.
7. Write the names of five 'Environmental Regulations Acts'?
8. Discuss the problems of Technology Transfers.

**GROUP: C** (*Answer any three questions*) **(15 x 3 = 45)**

9. Discuss the concept of value crisis in contemporary Indian society.
10. What is the concept of 'Club of Rome'? Discuss with suitable diagram.
11. Justify your view on inclusion of 'Professional Values and Ethics' course in the Engineering domain.
12. Discuss the Appropriate Technology Movement of Schumacher.
13. What do you understand by 'Sustainable Development'? How is 'Sustainable Development' linked with 'Energy Crisis & Renewable Energy Sources'?

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**ALIAH UNIVERSITY**
**Department of Civil Engineering**
**B.Tech/CENUGPE05/PEC-V/2024**
**Repairs and rehabilitation of structures**

**Time Allotted: 03 Hours** **Full Marks: 80**

- *The response to a complete question, along with any sub-questions, should be provided in one cohesive answer.*

## GROUP A
### (Short answer type questions)

**Answer all questions:** **(6×5 =30)**

1. (a) Discuss the cause of carbonation-induced corrosion of deterioration of concrete structures.
   (b) What are the causes of construction-related distress that occurs in concrete?
   (c) How w/c ratio affect the performance of a structure?
   (d) State the reason for the internal sulphate attack of concrete.
   (e) Explain the alkali aggregate reaction.
   (f) Differentiate between dry mix shotcrete and wet mix shotcrete.

## GROUP B
### (Long answer type questions)

**Answer any five of the following:** **(5×10 = 50)**

2. What is the Pull-Off test? Discuss the loading condition of this test. Discuss the procedure of the Pull-Off test.

3. (a) Define polymer mortar and micro concrete.
   (b) What are the common treatments used for crack stitching?

4. (a) Briefly discuss the distress due to weathering effects in concrete.
   (b) State the causes of distress due to natural calamities.

5. Explain the following: (a) Foundation-related distress. (b) Design /Load associated distress

6. (a) Discuss the causes of acid attack in concrete.
   (b) How can we resist acid attack in concrete?

7. (a) Discuss the benefits of Retrofitting and Rehabilitation.
   (b) Briefly describe the following terminologies in the context of repair and retrofitting of structures: Durability, Inspection, Evaluation, Restoration, Retrofitting, and Rehabilitation.



$CaCO_3$ $\cancel{Ca}$ $CO_2 + H_2O \rightarrow H_2CO_3$

$H_2CO_3 + Ca(OH)_2 \rightarrow CaCO_3 + H_2O$